

UNIVERSITI TUNKU ABDUL RAHMAN
LEE KONG CHIAN FACULTY OF ENGINEERING AND SCIENCE
ACADEMIC YEAR 2025
Data Mining

Group Assignment1

Part 1 (EDA & Data Pre-Processing) - Due in week 7, Mar 24, 6pm, email (report & Colab Notebook) to suls@utra.edu.my)

Part 2 and 3 (Models, evaluations & Deployment) - Due in week 12 (April 28, 6pm, email (report & Colab Notebook) to suls@utar.edu.my)

This assignment is worth 30% of total course marks

Sample datasets:

There are 1,027 observations in the **Heart1Assign.csv** file.

There are 10 new patients to be processed in **HeartNewPatients.csv** file.

Assignment Scenario:

The approach I adopt for the assignment is a realistic one... like a real life analytics company. You and your team are my employees. I have given you a task with minimal instructions on what to do and why. You need to think what is relevant in the context of the task , and what is not relevant. You will be penalized if you give me everything without considering what is relevant and what is not.

You and your team need to think and find out what are the steps to take, the sequence of those steps and why. I expect these in the report you submit.

Ultimately, determine if each of the 10-20 new patients (in the **HeartNewPatients.csv** file) should is predicted to have heart condition/ problem.

Deliverables:

I need the following in your submission to suls@utar.edu.my :

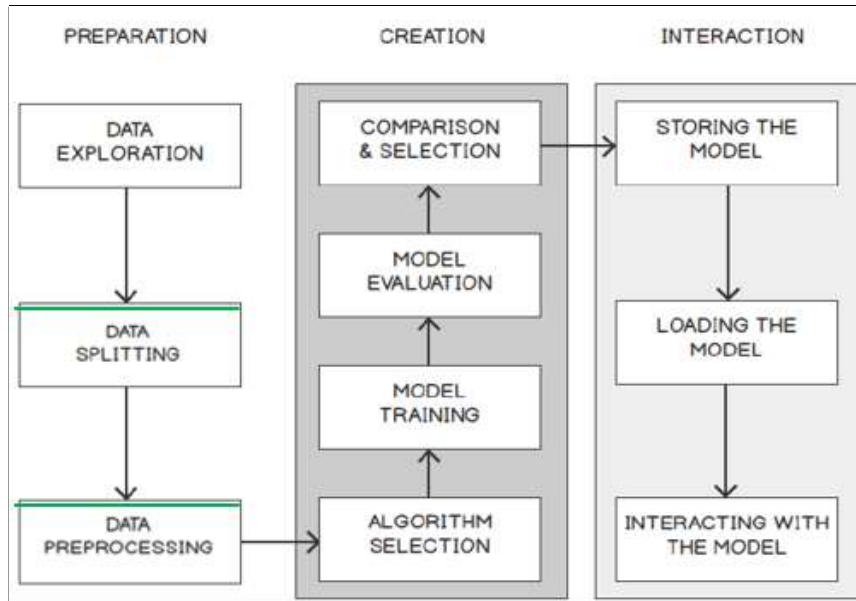
1. A report on the description and outputs, findings, experiences, opinions, remarks,...of tasks you and your team performed.
 2. A video recording presentation (of about 10 – 15 mins).
 3. The complete python program(s) in COLAB...codes...(as I will run your codes)
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Please, name your submission report as :

(<programme>) **ASSIGN1** <name of team lead> <student.id of team lead>

Example → (SE) Assign1 Student-name 1234567

The names and student ids of the team members must appear in the report.



The Stages of Comprehensive Machine Learning Project above serves as a guide to you.
